Research Proposal

Early Detection of Harmful Emergent behaviors in Fine-Tuned LLMs

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1 Objectives

Recent developments in LLMs have made them highly effective in complex reasoning, machine translation, conversational agents, content generation, and question answering. These capabilities are typically achieved by adapting base models to the tasks via post-training and finetuning methods. However, such adaptations introduce new challenges. Even when the finetuning data is benign, they can unexpectedly introduce harmful behaviors, increase the likelihood of unsafe responses, or destabilize long-chain reasoning, so it is important to know what and when certain behaviors emerge. In current practice, these failures are often discovered only after a model has been trained and integrated into evaluation or product pipelines. For large-scale pre-training or full-model post-training, correcting these problems is extremely costly, and even for cheaper finetunes such as LoRA, the sheer number of runs and deployment contexts makes late detection impractical. Because pre-training is expensive and difficult to instrument in detail, finetuning can be a practical proxy setting in which to study when and how risky behaviors emerge, to develop techniques that can eventually transfer back to the larger training domain.

The main question to answer here is:

> **Can we detect and understand the early emergence of harmful behaviors during finetuning and take action before these behaviors fully emerge in model outputs?**

To answer this, we will focus on two main ideas:

- **Self-explaining internal states:** Train a self-explanation model that can understand internal activations and simple trajectory statistics and convert it into natural language descriptions of behavioral shifts, which makes understanding the model's inner workings easier. We are often aware of the shifts we care about, but in this part, we want to spot when the model is drifting in ways we did not intend.
- **Token entanglement and harm geometry:** Model how benign and harmful concepts become intertwined in representation space, and use this entanglement as a mechanistic and predictive signal for harmful emergence.

Additionally we wi,ll create a model that can combine these signals into an early-warning system able to explain and guide interventions during finetuning.

2 Background

Although LLMs have been empirically successful in many domains, they still exhibit different kinds of failures and vulnerabilities. For example, they are widely known to be vulnerable to hallucinations and behavioral shifts under small changes in data or prompts [He et al., 2024].

Benchmarks such as TruthfulQA show that larger models can sometimes be *less* truthful than smaller ones [Lin et al., 2022], and repeated finetuning across domains amplifies these risks.

Mechanistic interpretability has made progress in opening the "black box" of deep models. It helps identify concrete circuits such as induction heads, analyze attention patterns while also allowing us to find the features responsible for specific behaviors [Elhage et al., 2021, Olsson et al., 2022]. More recent work has developed model diffing methods such as crosscoders, which learn a shared dictionary of latent directions in a base model and its fine-tuned variant and use these to track how concepts shift or newly appear during fine-tuning[Lindsey et al., 2024, Minder et al., 2025]. This line of work shows that it is possible to attribute fine-tuning-induced behavioral changes to concrete internal features. However, most work is still post-hoc and offline, meaning that they analyze internal mechanisms after applying changes to the model and do not provide online or predictive tools that can help us detect instability in reasoning or harmful behaviors that might emerge during finetuning.

One way to approach this is by using explaining models. Recent work on explaining models shows that a model can map its internal states to natural-language descriptions of what the model is doing, suggesting a route to interpretable online monitoring Li et al. [2025]. At the same time, representation geometry suggests that harmful and benign concepts may become entangled by sharing embedding neighborhoods or features in a way that can cause gradient updates on benign data to push representations towards harm-relevant directions. For example, the embedding for the word "food" can be close to the embedding of the word "poison".

3 Research Plan

3.1 Self-explaining internal states

Explainer model. Building on the framework of Li et al. [2025], we will treat each fine-tuned checkpoint along the training trajectory and train an explainer model E that learns to talk about what is going on inside it. In Li et al., the model is given access to its own internal features together with a question about that feature (e.g., "what pattern does this feature detect?" or "what happens if we patch this activation?"). A separate automated interpretability pipeline produces supervision that the model is trained on.

We adapt this idea to the finetuning setting. Since finetuning happens over time, We will define checkpoints at which we save the model's internal state and trajectory. We look into different things such as:

- Activations from a small number of layers.
- Trajectory statistics like LoRA weight differences relative to the base model or large gradient spikes.
- harm space statistics. In this part, we analyze how strongly activations project onto the harm subspace $\mathcal{H}$, which is defined below.

We then combine these internal signals with a short question E is trained to answer in natural language, describing the model's behaviors at that point in finetuning. For example, it might be becoming more willing to answer borderline questions.

Online monitoring. Then, during finetuning, every K steps we will compute the internal summary for the current checkpoint and query E with a small set of fixed questions about the behavior. The result from E will be mapped to a small set of behavioral scores and short explanations. The goal is that E not only predicts whether the behavior is drifting in a dangerous direction, but also gives us an intuition about the reason of this behavior being tied to internal changes. For example, "representations of benign medical questions are becoming more similar to self-harm questions".

Forecasting future internal states. Additionally, instead of describing each checkpoint, we will also extend this to future checkpoints by combining it with recent work that shows it is possible to predict weight trajectories from partial training sequences Shou et al. [2025]. In this direction, a small model will be trained on historical finetuning runs to take as input a prefix of LoRA weight differences and basic training statistics, and to predict the next few checkpoints in the trajectory. We can then use this model to forecast future internal summaries, and also feed these predicted summaries into the self-explainer E. We can then ask E to produce natural language explanations of how the model is likely to behave if training continues.

3.2 Token entanglement and harm-space monitoring

In this part the goal is to model a harm-relevant subspace and token entanglement, with both feeding into a practical early-warning system for finetuning.

Harm space construction. We will construct a "harm space" $\mathcal{H}$ in embedding or feature space by collecting representations of harmful tokens and extracting a subspace that captures the main harm directions.

Entanglement metrics. Then for each token and selected internal activations, we define an entanglement score based on projection or similarity to $\mathcal{H}$ and a time series of these scores over finetuning steps for a fixed set of prompts. This quantifies exactly how much benign tokens and internal states are drifting towards harmfulness during finetuning. Even if token entanglement itself does not turn out to be the main driver, these harm-space projections are still valid and useful as a general geometric early-warning signal.

Testing the token-entanglement hypothesis. In addition to using harm-space drift directly, we will also test the more specific hypothesis that token entanglement can cause the emergence of harmful behavior during finetuning. To do this and see whether entanglement is mechanistically important we will first identify benign tokens with highest entanglement scores and then run finetuning variants where these tokens are removed or the loss is masked with respect to these tokens. If changes in harmful behavior track these manipulations, this would support a causal role for entanglement.

3.3 Integrated early-warning system

Finally, we can go on to train a simple risk model F that combines the different signals collected during finetuning. F will take as input the behavioral scores and explanations derived from E's self-explanations. It can also take in the harm-space and entanglement features.

F outputs a harmfulness score for the current checkpoint and tell us if we need to intervene. Additionally, E can also be prompted to "explain" high-risk predictions by describing which internal changes and entanglement patterns are likely responsible. Together, this gives both a numerical and a human-readable data about what is going wrong inside the model while it is being finetuned.

4 Significance

Finetuned LLMs are highly used in many domains, however the processes by which updates lead to harmful emergent behaviors are poorly understood and largely unmonitored. This project advances this safety through two main contributions:

- an explaining monitoring layer, inspired by Li et al. [2025], that uses internal activations, weight-diffs and trajectory summaries to produce natural-language descriptions and scores for behavioral shifts.
- a token entanglement and harm-space framework that links representation to harmful emergence.

Together, these components form a mechanistic early-warning system for finetuning which detects risky runs and allow us to adjust before harmful behaviors fully emerge. In the longer term, such tools could reshape deployment practice by making safety checks an integral, automated part of the training loop rather than a purely post-hoc evaluation step.

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